# Foreign Exchange Market

# Exchange Rates

- Exchange rate measures the value of one currency in units of another currency.
- The rates at which spot and forward transactions take place are known as "spot rate" and "forward rate" respectively
- Nominal exchange rate (NER) and the real exchange rate(RER)
- RER indicates the real purchasing power of one currency relative to another currency; it is the NER adjusted for changes in the relative purchasing power of each currency since some base period.

$$REET = NEER \times [(1+ INF_f)/ (1+INF_h)]$$

foreign country inflation rate

home country's inflation rate

Real interest rate = Nominal rate - Inflation

→ NEER is weighted average of bilateral nominal exchange rates of home currency against selected foreign currencies. REER → obtained by deflating NEER by corresponding inflation rates

# Exchange Rates Cont…

- In order to obtain a measure of the multilateral ER, economists have developed the concepts and measures of the "**Nominal Effective Exchange Rate**" (NEER) and the "**Real Effective Exchange Rate**" (REER).
- NEER is a weighted average of the bilateral nominal exchange rates of the home currency against selected foreign currencies, REER is obtained by deflating NEER by the corresponding relative inflation rates.
- Weights are given on the basis of trade.

# Direct Vs. Indirect Exchange Rate Quotations

- There is a practice of giving either number of home currency units, say rupees, per unit of foreign currency, say US Dollar or the number of units of foreign currency, say Dollar per unit of home currency, say a rupee, when banks deal with non-bank customers.
- When there is no transaction cost, direct exchange rate between any two currencies e.g. rupee and euro is exactly equal to the implicit indirect exchange rate via US dollar.

# Cross Exchange Rate

- It represents the relationship between two currencies that are different from one's base currency. In India the cross reference refers to the relationship between two non rupee currencies.
- It represents the exchange rate between two currencies via another currency
- Let spot exchange rate between rupee and pound as: S (₹/ £)
- The cross exchange rate = S(₹/ \$) × S( \$/ £)

# Arbitrage

- Arbitrage can be defined as capitalizing on a discrepancy in quoted prices to make a riskless profit
  - Triangular arbitrage
  - Locational arbitrage

# Triangular Arbitrage

Triangular arbitrage is possible when a cross exchange rate quote differs from the rate calculated from spot rate quotes.

Example: Suppose we have Rs. 700 we have to convert it to pound. 1$ = ₹70, 1£= ₹90, 1£= $1.2

Direct: Rupee to pound = ₹700 / 90 = £7.77

→ Assuming no transaction costs

Indirect: Rupee to Dollar = ₹ 700/ 70 = $10

Dollar to Pound = $10 /1.2 = £8.33

# Example:

Suppose we have Rs. 700 we have to convert it to pound. 1$ = ₹70, 1£= ₹98, 1£= $1.4

Direct: Rupee to pound = ₹700 / 98 = £7.142857

Indirect: Rupee to Dollar = ₹ 700/ 70 = $10

Dollar to Pound = $10 /1.4 = £7.142857

(No arbitrage)

# Locational Arbitrage

- Locational arbitrage is possible when a bank's buying price (bid price) is higher than another bank's selling price (ask price) for the same currency.

<u>Example</u>

| *Bank A* | **Bid** | **Ask** | ***Bank B*** | **Bid** | **Ask** |
|---|---|---|---|---|---|
| **Indian(₹)** | **$0.015** | **$0.017** | **Indian(₹)** | **$0.019** | **$0.020** |

Buy **Indian(₹)** from Bank A @ $0.017, and sell it to Bank B @ $0.019.
Profit = $.002/ Indian ₹.

# Interest Rate Parity (IRP)

→ Interest rate differential b/w two countries is equal to diff. b/w forward exchange rate & spot rate

- The forward rate differs from the spot rate by an amount that sufficiently offsets the interest rate differential between two currencies.

$$\text{forward premium} = \frac{(1 + \text{home interest rate})}{(1 + \text{foreign interest rate})} - 1$$

- The IRP relationship can be rewritten as follows:

- $FP = \frac{F - S}{S} = \frac{(1+i_H)}{(1+i_F)} - 1 = \frac{(i_H - i_F)}{(1+i_F)}$

- FP = forward premium, F= forward rate in home currency, S = spot rate in home currency, i(h) = home interest rate, i(f) = foreign interest rate, F = S (1+FP)

- The approximated form, $Fp \approx i_H - i_F$, provides a reasonable estimate when the interest rate differential is small.

# Exchange Rate Movement

- When a currency declines in value, it is said to depreciate. When it increases in value, it is said to appreciate.
- The percentage change in the value of a foreign currency is computed as

Reference pt. is foreign currency

$$\frac{S_t - S_{t-1}}{S_{t-1}}$$

Depreciation - done by market forces (Market determined)

Devaluation - done by regulatory body intentionally due to economic factors

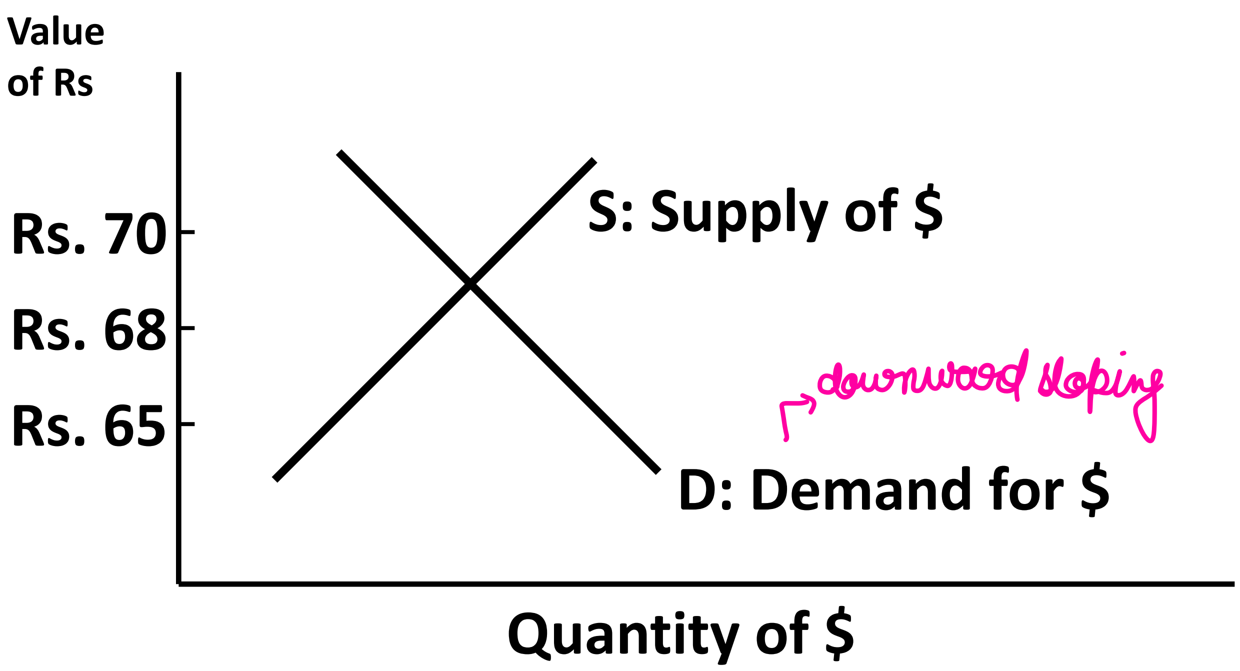
where $S_t$ denotes the spot rate at time $t$

*Positive* % change represents appreciation of the foreign currency, while a negative % change represents depreciation.

home currency depreciating

# Exchange Rate Determination

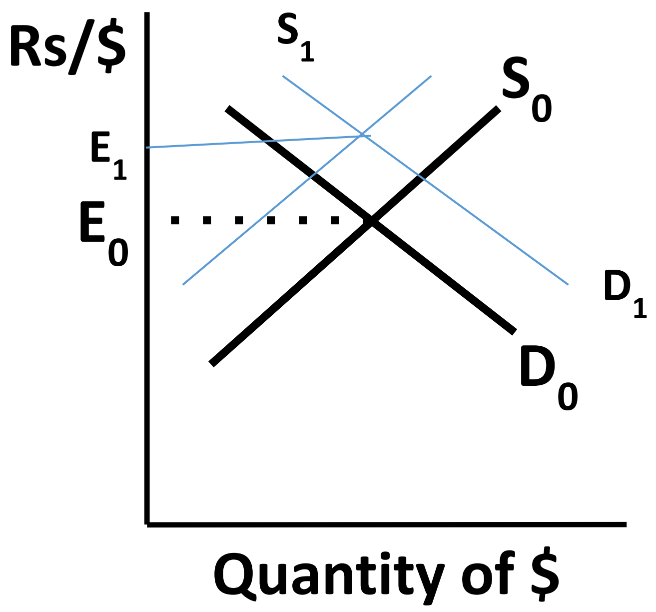
equilibrium exchange rate determined by supply & demand




- The exchange rate represents the price of a currency, or the rate at which one currency can be exchanged for another.
- Demand for a currency increases when the value of the currency decreases, leading to a downward sloping demand schedule.
- Supply of a currency increases when the value of the currency increases, leading to an upward sloping supply schedule.
- Equilibrium equates the quantity of currency demanded with the supply of currency for sale.
- In liquid spot markets, exchange rates are not highly sensitive to large currency transactions.

→ Relative word important here

# Relative Inflation Rate




Indian inflation rate↑

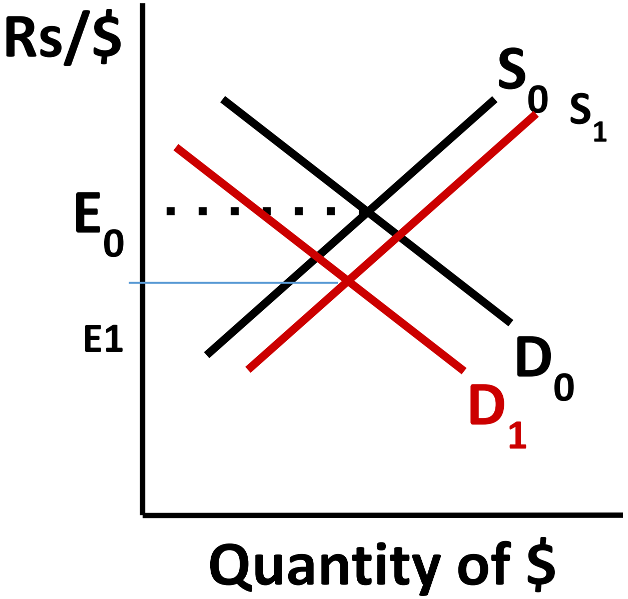
→ Indian commodities will become more expensive

⇒ ↑ Indian demand for US goods hence $

⇒ ↓ US desire for Indian goods, and hence the supply of $.

↳ America won't buy Indian commodities, so supply of $ ↓

# Relative Interest Rates




Indian interest rate↑

↳ eg Indian banks giving more interest

⇒ ↓Indian demand for US bank deposits and hence $

⇒ ↑ US desire for Indian bank deposits, and hence the supply of $.

# Relative Interest Rates

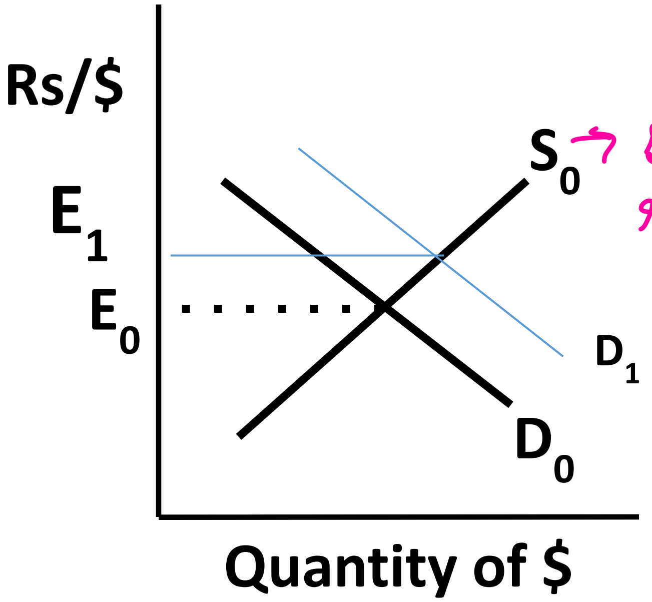
- A relatively high interest rate may actually reflect expectations of relatively high inflation, which discourages foreign investment.
- Therefore, consider real interest rates, which adjust the nominal interest rates for inflation.
- Real interest rate = Nominal interest rate – Inflation rate
- This relationship is sometimes called the Fisher effect.

always consider real interest rate

↓

adjusted by inflation

# Relative Income Level




supply curve will remain same

Indian income level ↑

more luxury product demand by Indians

- ⇒ ↑ Indian demand for US goods, and hence $.
- ⇒ No expected change for the supply of $.

supply curve will remain same

# Other factors

- Foreign trade barriers, → tariffs, import duties [current account balance affected]
- Intervention of foreign exchange market → RBI can intervene to manage sudden changes
- Expectations
  - News impact → coming randomly, can be both +ve & -ve news
  - Institutional investors often take currency positions based on anticipated interest rate movements in various countries.

# Exchange Rate System

- Fixed → not market determined, fixed by the gov. like China
- Floating → purely market driven
- Managed → market driven, but if any situation arises RBI can intervene to manage fluctuations

# Foreign Exchange Market

- The market in which national monetary units or claims are exchanged for the foreign monetary units is defined as foreign exchange market (FEM).
- The FEM is not a physical place; it is an informal, electronically linked network of big banks, foreign exchange brokers and dealers whose function is to bring buyers and sellers together. → just like stock exchange
- The trading in FEM is usually done 24 hours a day by telephones, → due to time difference display monitors, telex and fax machines, and the satellite communications network called the *Society for Worldwide International Financial Telecommunications (SWIFT)*, which is a computer- based communication system.

# Foreign Exchange Market Cont…

- Most foreign exchange markets in developing countries are either pure dealer markets or a combination of dealer and auction markets.
- In the dealer markets, some dealers become market makers and play a central role in the determination of exchange rates in flexible exchange rate regimes.

bigger stakeholders
play imp role in
determining rates

- Market makers set two-way exchange rates at which they are willing to deal with other dealers.
- In auction markets, an auctioneer or auction mechanism allocates foreign exchange by matching supply and demand orders.
- In pure auction markets, order imbalances are cleared only by exchange rate adjustments.

# Foreign Exchange Market Cont…

open bid → both buying & selling

double auction - both bank

- Two levels of FEM
  - Direct interbank level → done through telephone mostly
  - It has been sometimes characterised as a "**decentralised, continuous, open-bid, double-auction**" market.
- Indirect level via foreign exchange broker
  - The banks put orders with brokers who put them on "books", and try to match purchases and sales orders for different currencies. They charge commission to both the buyers and sellers.
  - This market is characterised as **"quasi-centralised, continuous, limit-book, single-auction"** market.

# Foreign Exchange Market Cont…

- **Retail Market**
  - The exchange of bank notes, bank drafts, currency, ordinary and traveller's cheques between private customers, tourists and banks takes place in the retail market
- **Wholesale Market**
  - The wholesale market is primarily an inter-bank market in which major banks trade in currencies held in different currency-dominated bank accounts i.e., they transfer bank deposits from seller's to buyer's accounts.

# Retail Vs. Interbank Spot Rate

- Exchange rate between banks is determined by interbank market
- Exchange rate charged on banks' clients is based on interbank rate
- Banks charge their customers more than the interbank selling or ask rate and pay their customers less than the interbank buying or bid rate

Interbank rate
↑
Ex If 1 pound = 90 Rs
92 Rs → charge
88 Rs → give

Banks → charge their customer more than interbank selling rate (ask rate)
→ pay " " less " " buying rate (bid rate)

# Foreign Exchange Market Cont…

- The major sources of supply of foreign exchange in the Indian foreign exchange market are receipts on account of exports and invisibles in the current account and inflows in the capital account such as foreign direct investment (FDI), portfolio investment, external commercial borrowings (ECB) and non-resident deposits.
- On the other hand, the demand for foreign exchange emanates from imports and invisible payments in the current account, amortisation of ECB (including short-term trade credits) and external aid, redemption of NRI deposits and outflows on account of direct and portfolio investment

# Players in the Foreign Exchange Market in India

- Reserve Bank of India (major stakeholder)
- ADs, mostly banks who are authorised to deal in foreign exchange, foreign exchange brokers who act as intermediaries, and customers—individuals, corporates
- All scheduled commercial banks, which include public sector banks, private sector banks and foreign banks operating in India, belong to category I of ADs. All upgraded full-fledged money changers (FFMCs) and select regional rural banks (RRBs) and co-operative banks belong to category II of ADs. Selected financial institutions such as EXIM Bank belong to category III of ADs.

Three type of authorized dealers

Category I: Commercial banks – public sector banks, private sector banks, foreign banks

Category II: Full-fledged money changers (FFMCs), rural banks (RRBs), cooperative banks

Category III: Selected financial institutions such as EXIM bank

# Players in the Foreign Exchange Market in India

public sector units, corporates and business entities → customer segment

- The customer segment of the foreign exchange market comprises major public sector units, corporates and business entities and dominated by select large public sector units such as Indian Oil Corporation, ONGC, BHEL, SAIL, Maruti Udyog and also the Government of India (for defence and civil debt service), private companies like Reliance Group, Tata Group and Larsen and Toubro, among others.

like ONGC, Indian oil, BHEL, SAIL, Maruti, gov. of India, Reliance group, Tata group, L&T group

- In recent years, foreign institutional investors (FIIs) have emerged as major players in the foreign exchange market.

regulatory body

- Foreign Exchange Dealers' Association of India (FEDAI) plays a special role in the foreign exchange market for ensuring smooth and speedy growth of the foreign exchange market in all its aspects.

FIIs → come for stock market but have to go through foreign exchange market

# Foreign Exchange Market Trading

- In the Indian foreign exchange market, spot trading takes place on four platforms, viz., FX CLEAR of the CCIL set up in August 2003, FX Direct that is a foreign exchange trading platform launched by IBS Forex (P) Ltd. in 2002 in collaboration with Financial Technologies (India) Ltd., and two other platforms by the Reuters - D2 platform and the Reuters market data system (RMDS) trading platform that have a minimum trading amount limit of US $ 1 million. → high net worth individuals & companies only can participate
- These trading platforms cover the US dollar-Indian Rupee (USDINR) transactions and transactions in major cross currencies (EURIUSD, USDIJPY, GBPIUSD etc.), though USD-INR constitutes the most of the foreign exchange transactions in terms of value. It is the FX-CLEAR of the CCIL that remains the most widely used trading platform in India. This

# Objectives of Exchange Rate Policy in India

- To ensure that economic fundamentals are reflected in the external value of the rupee.
- To reduce excess volatility in exchange rates, and to ensure that the market correction of overvalued or undervalued exchange rate is orderly and calibrated.
- To help maintain an adequate level of foreign exchange reserves.
- To help eliminate market constraints in the way of development of a healthy FEM.

# Central Bank Intervention in FEM

- To influence trend movements in the exchange rates because they perceive long-run equilibrium values to be different from actual values
- To maintain export competitiveness
- To manage volatility to reduce risks in financial markets
- To protect the currency from speculative attack and crisis

MS has been sterilised in the domestic market

net affect = 0

# Types of Intervention

whenever central bank tries to buy or sell foreign currency → If RBI buys foreign currency → MS↑, inflation↑

↓

so what central bank does

Buy foreign currency

⇓

Foreign assets ↑

MS↑, sell bonds to public, to decrease MS in economy

- *Sterilised* intervention occurs when the purchase or sale of foreign currency is offset by a corresponding sale or purchase of domestic government debt to eliminate the effects on domestic money supply.

- *Non-sterilised* intervention occurs when the authorities purchase or sell foreign exchange, normally against their own currency, without such offsetting actions.

simply buy or sell foreign currency → don't control MS in the economy

# Types of Intervention Cont…

| *Intervention* | *Effects on NFA* | *Effects on NDA* | *Effects on M* |
|---|---|---|---|
| Non-sterilised foreign exchange | | | |
| Intervention (Purchase) | + | 0 | + |
| Sterilised foreign exchange | | | |
| Intervention (Purchase) | + | – | 0 |
| Non-sterilised foreign exchange | | | |
| Intervention (Sale) | – | 0 | – |
| Sterilised foreign exchange | | | |
| Intervention (Sale) | – | + | 0 |

→ freedom to convert local assets into foreign assets & vice versa

# Currency Convertibility

- It refers to the freedom to convert local financial assets into foreign financial assets and vice versa at market-determined rates of exchange
- It is associated with changes of ownership in foreign / domestic financial assets and liabilities and embodies the creation and liquidation of claims on, or by, the rest of the world

# Partial Vs. Full Convertibility

100% allowed to convert local to foreign assets & vice versa

- Complete freedom to convert local financial assets into foreign financial assets and vice versa at market-determined rates of exchange
- The partial convertibility would mean that the freedom to buy or sell currency is only limited in amount; or it is only for the foreigners.

  certain restrictions
- According to some people, full convertibility means removing trade and exchange controls completely, and shifting away from the fixed, managed, pegged ER systems towards the flexible or floating ER system

# Characteristics of currency convertability

- Freeing the ER regime (market determined exchange rate)
- Eliminating import licensing, custom duties, import taxes and tariffs, advance import deposits, export incentives, multiple exchange rates (ER is complete free)
- Removing restrictions on international services transactions, on earning, availability, use, retention, and holding of foreign ex change at home and abroad, and on international capital movements as well as buying and selling of foreign exchange (No restrictions)
- Giving freedom to remit abroad profits, dividends, and any other legitimate income in foreign currencies.

# Prerequisites to make this system convertible

- The ER must be realistic
- The country should enjoy low inflation rate and internal financial stability
- Foreign exchange reserves should be large in practice
- The trading partners should open up their trade and payments systems
- Debt levels, particularly external debt level, should be low
- Fiscal and monetary austerity, prudence, consolidation, and drastic reduction in fiscal deficit should be achieved
- Labour market reforms, including unemployment insurance, job retraining, and wage discipline should be achieved

1. ER must be realistic
2. low inflation rate
3. financial stability
4. Low Debt levels
5. ForEx reserves should be high
6. Fiscal deficit should be low

# Dangers of CC

- It increases the risk of capital flights both ways, which increases the volatility in exchange rates and financial markets. This risk is very real in practice due to the fact that 90 per cent of transactions in FEMs are not related to trade.
- It tends to increase unemployment, and reduce output and wages.
- It may accentuate inflationary pressures because foreign goods become available at higher prices
- It can result in the flooding of the domestic markets with imports, particularly non-essential ones
- It increases the misuse of foreign exchange not only for luxury and leisure industry but also for smuggling of goods, drugs, arms, and for other nefarious activities
- Unlimited access to short-term external borrowings, and giving unrestricted freedom to domestic residents to convert their domestic bank deposits and idle assets in response to market developments or exchange rate expectations would cause extreme domestic financial vulnerability as the experience of the Asian crisis has taught us.

# Foreign Exchange Reserve

- The main objectives in managing a stock of reserves for any developing country are preserving their long-term value in terms of purchasing power over goods and services, and minimising risk and volatility in returns.
- The desirable size of reserves can be explained mainly by the factors: (i) the size of the economy, (ii) its vulnerability to the current and capital accounts, (iii) exchange rate flexibility, (iv) opportunity cost (v) financial market integration etc.
- Sources of accretion to foreign exchange reserves are: Foreign Investments, NRI Deposits, External Assistance, ECBs etc.

↳ External commercial borrowings

# Foreign Exchange Reserve Cont….

- High level of FERs in India reflects
  - The success of reforms
  - India's ability to meet financial obligations and maintain the country's monetary stability
  - India's ability to import more goods, her ability to absorb the shocks and uncertainties in the world economy
  - Ability to cope up with crisis and capital flight
  - Greater availability of liquidity and confidence
  - Higher sovereign rating, greater backing for domestic currency, and finer terms on debt

# Foreign Exchange Reserve Cont….

- Adverse implications of very high level of FERs
  - Maintenance or holding of excessive FERs is costly. The cost of holding FERs is the opportunity cost of investing them in productive activities.
  - The accumulation of FERs augments the domestic money supply, and, in order to avoid excessive growth in money supply, the RBI has to conduct sterilisation operations by selling the government securities.
  - High FERs have increased the interest burden of the economy, because a large part of them are in the form of NRI deposits.
  - Increase the vulnerability in the market.
  - The reserves have not been built up due to favourable balance of trade or surplus on the current account.

# Foreign Exchange Reserve Cont….

- Indicators of Adequacy of Foreign Exchange Reserves are:
  - Import cover of Reserves (Months),
  - FER to Reserve Money
  - FER to Broad Money
  - FER to External Debt
  - FER to Short-term Debt
  - FER to GDP

# Importance of Foreign Capital

- To relax the domestic savings constraint
- To help to overcome the foreign exchange barrier
- To provide access to the superior technology, superior managerial skills, and bigger markets
- To provide risk-sharing capital financing
- To furnish the funds needed for the full utilisation of the existing production capacity
- To promote efficiency and productivity through international competition.

# Determinants of Foreign Capital

- Interest rate
- Degree of openness
- Legal and institutional structure
- Relative rates of inflation
- The relative stability of the exchange rates
- Business cycle
- Changes in the preferences of the investors

# Components of Foreign Capital in India

- Debt
- External Commercial Borrowings
- Loans from IMF
- NRI Deposits
- FDI
- FII

# FDI

- Foreign direct investment (FDI) happens when a firm invests directly in facilities in a foreign country
- Involves ownership of entity abroad for
  - production
  - Marketing/service
  - R&D
  - Raw materials or other resource access
- The degree of direct managerial control depends on the extent of ownership of the foreign entity and on other contractual terms of the FDI
- A firm that engages in FDI becomes a multinational enterprise

# Types of FDI

- Acquisitions
  - Purchase an existing company in the foreign country
- Greenfield Investments
  - Set up a new company “from the ground up” in the foreign country
  - Example: Motorola investments money in China and builds a new plant to produce cell phones
- Wholly Owned Subsidiary
  - Occurs when the company in the foreign country is entirely controlled/owned by one single company.
- Joint Ventures
  - Occurs when two or more companies together form a new company in the host country
  - Example: Maruti Suzuki

# Horizontal Vs. Vertical FDI

- Horizontal
  - Investment in the “same” industry as a firm operates in at home
  - Example: Macdonald, Starbucks etc.
- Vertical
  - Investment in a downstream supplier (backward) or upstream purchaser (forward) as compared to the business that the firm operates in its home country
  - Backward: One input for any product is produced in another country
  - Forward: Dealers acquisition to sell the product in other country

# Impact of FDI on Host Country

- Resource Transfer
  - Capital
  - Technology
  - Management
- Employment
- Balance of Payment effect
- Effect on Competition

# Entry Routes to India

- Automatic Route
  - No prior permission required
  - Only information to the Reserve Bank of India within 30 days of inflow
- Prior Permission from Foreign Investment Promotion Board

# Foreign Institutional Investments

- It is defined as the investment made by a foreign investor in the shares of company that is listed in hosting country or in bonds offered by the hosting country. For example if a foreign investor buys the share of Infosys that qualifies as FII investments.
- FIIs may invest in:
    - Securities in the primary and secondary equity markets
    - Units issued by domestic mutual funds
    - Dated Government securities
    - Derivatives traded on a recognized stock exchange
    - Commercial papers etc.

# Who can be FII?

- An institution established or incorporated outside India as a pension fund, mutual fund, investment trust, insurance company, or reinsurance company
- An international or multilateral organization or an agency thereof, or a foreign governmental agency, sovereign wealth fund, or a foreign central bank
- An asset management company, investment manager or advisor, bank, or institutional portfolio manager that is established or incorporated outside India and proposes to make investments in India on behalf of broad-based funds and its proprietary funds, if any
- Trustee of a trust established outside India who proposes to make investments in India on behalf of broad-based funds and its proprietary funds, if any; v. University funds, endowments, foundations, charitable trusts, or charitable societies.

# Investment Through Sub-Accounts

- Sub-account refers to any person who is resident outside India on whose behalf investments are proposed to be made in India by a foreign institutional investor, and who is registered as a sub-account under the SEBI (FII) Regulations, 1995.
- Who can hold the sub account?
  - Proprietary fund of a registered foreign institutional investor.
  - Foreign individual who has a net worth of not less than US $ 50 million, holds a valid passport of a foreign country for a period of at least five years, holds a certifiate of good standing from a bank, and is the client of the FII for a period of at least three years
  - Foreign corporate that has its securities listed on a stock exchange outside India, having an asset base of not less than US $ 2 billion and having an average net profit of not less than US $ 50 million during the three fianancial years preceding the date of application.

# FII Limits

- The total holding of each FII/sub-account under this scheme should not exceed 10 percent of the total paid up capital or 10 percent of the paid-up value of each series of convertible debentures issued by the Indian company
- The total holding of all the FIIs/sub-accounts put together should not exceed 24 percent of the paid-up capital or the paid-up value of each series of convertible debentures. This limit of 24 percent can be increased to the sectoral cap/statutory limit as applicable to the Indian company concerned, by passing a resolution of its Board of Directors, followed by a special resolution to that effect by its General Body.

# Difference between FDI and FII

**FDI**

- Applicable to many sectors
- Long term investments
- Investment in fixed capitals
- Investments are stable
- Targets specific company
- Transfer of resources in many forms like technology, strategy etc.
- Entry and exit difficult

**FII**

- Confined to Financial Sector
- Short-term
- Only financial capital
- Highly volatile
- Whole financial market
- Transfer of funds only
- Entry and exit easy